

IMPHAL, India

WMO: 426230

Lat: 24.760N

Lon: 93.897E

Elev: 774

StdP: 92.36

Time Zone: 5.50 (IND)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	6.4	3.4	5.3	10.0	4.4	5.7	9.9	4.6	19.3	4.0	19.2	0.1	140	0.256

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	32.2	24.0	31.5	24.0	30.9	23.9	26.1	30.3	25.6	29.7	25.2	29.1	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	22.1	28.8	24.5	21.4	28.1	24.1	20.9	27.6	86.9	30.6	84.6	30.0	82.7	29.3	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.4	3.5	3.0	34.2	0.8	0.8	2.4	34.8	1.9	35.2	1.4	35.7	0.8	36.2		
			2.7	27.6	1.0	0.9	2.0	28.2	1.5	28.8	0.9	29.3	0.2	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	14.8	17.1	20.5	23.0	24.6	26.0	26.1	26.2	25.8	23.6	19.1	15.4
	DBStd	4.49	1.61	2.09	2.05	2.26	1.73	1.38	1.14	1.12	1.13	1.98	1.90	1.81
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	268	111	45	6	2	0	0	0	0	0	1	12	92
	CDD10.0	4330	148	198	325	389	451	479	501	502	474	422	274	167
	CDD18.3	1557	1	10	72	141	193	229	242	244	224	164	36	1
	CDH23.3	11922	38	208	761	1156	1421	1781	1833	1849	1553	1011	281	30
	CDH26.7	3299	0	22	205	397	451	521	456	487	448	282	31	1
Wind	WSAvg	0.9	0.6	0.9	1.2	1.2	1.0	1.0	1.1	1.2	0.9	0.7	0.6	0.6
Precipitation	PrecAvg	2067	15	48	81	145	260	359	375	314	247	145	53	16
	PrecMax	2895	53	227	208	401	462	489	514	512	399	267	142	80
	PrecMin	1197	3	13	10	38	117	203	237	204	141	44	21	0
	PrecStd	328	14	46	54	83	78	67	77	82	69	60	36	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	28.9	31.8	33.6	33.0	32.7	32.0	32.5	32.2	31.7	29.0	25.6
		MCWB	16.0	17.0	18.9	19.6	24.1	24.9	25.8	26.5	25.6	24.6	21.8	17.4
	2%	DB	24.4	27.2	30.2	31.8	31.4	31.6	31.0	31.3	31.4	30.7	27.5	24.1
		MCWB	15.4	16.6	18.6	20.9	23.4	24.7	25.3	25.9	25.1	23.9	20.3	16.6
	5%	DB	23.2	25.8	29.0	30.4	30.5	30.6	30.1	30.4	30.4	29.6	26.3	22.9
		MCWB	14.9	16.1	18.6	21.0	23.2	24.4	25.0	25.5	24.8	23.4	19.5	16.2
	10%	DB	21.8	24.5	27.5	29.1	29.3	29.6	29.2	29.4	29.3	28.3	25.1	21.6
		MCWB	14.4	15.6	18.2	21.1	22.9	24.2	24.7	25.1	24.5	22.8	18.7	15.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	19.1	21.5	23.7	25.2	26.1	26.3	27.3	26.3	25.7	22.4	18.6
		MCDB	22.9	25.2	27.8	28.8	30.6	30.9	30.6	31.2	31.1	30.0	27.7	23.3
	2%	WB	16.2	18.0	20.6	23.0	24.5	25.4	25.7	26.4	25.6	24.7	21.2	17.5
		MCDB	22.2	24.7	27.0	28.5	29.7	29.8	29.7	30.2	30.1	29.2	25.8	22.6
	5%	WB	15.4	17.1	19.8	22.4	24.0	25.0	25.2	25.7	25.1	24.0	20.5	16.6
		MCDB	21.7	23.8	26.6	27.7	28.8	29.1	29.0	29.4	29.3	28.4	24.5	21.8
	10%	WB	14.7	16.2	19.0	21.7	23.4	24.6	24.9	25.2	24.7	23.3	19.7	15.9
		MCDB	21.0	23.1	25.3	26.9	27.8	28.4	28.4	28.7	28.5	27.2	23.4	20.7
Mean Daily Temperature Range	5% DB	MDBR	16.4	16.3	14.6	11.3	9.2	7.0	6.2	6.5	7.5	9.8	13.6	14.8
		MCDBR	17.6	17.9	16.7	14.2	11.4	8.8	7.7	8.2	9.3	11.5	14.5	15.6
		MCWBR	9.2	8.5	6.9	5.6	4.8	3.5	3.1	3.9	4.0	5.2	7.6	8.6
	5% WB	MCDBR	16.5	16.0	13.8	11.1	9.7	7.5	6.9	7.6	8.3	9.7	12.2	14.4
		MCWBR	9.0	8.2	6.9	5.3	4.6	3.4	3.1	3.9	3.7	4.6	6.8	8.3
Clear-Sky Solar Irradiance	taub	0.393	0.438	0.592	0.572	0.547	0.498	0.435	0.446	0.454	0.423	0.370	0.356	
	taud	2.230	2.109	1.735	1.849	1.975	2.179	2.377	2.332	2.283	2.321	2.387	2.389	
	Ebn at Noon	853	838	727	756	772	804	856	849	835	845	876	882	
	Edh at Noon	125	151	231	210	185	150	123	128	132	121	106	103	
All-Sky Solar Radiation	RadAvg	4.14	4.82	5.20	5.27	4.94	4.36	4.17	4.33	4.29	4.26	4.24	3.83	
	RadStd	0.27	0.29	0.41	0.54	0.54	0.42	0.39	0.30	0.27	0.31	0.27	0.32	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page